

Analysis of NBA Player's Role Type: Using Mixture-of-Factors Model

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Data Description and Preprocessing

<https://www.kaggle.com/datasets/mattop/nba-draft-basketball-player-data-19892021>

RangeIndex: 1922 entries, 0 to 1921

Data columns (total 24 columns):

#	Column	Non-Null Count	Dtype
0	id	1922 non-null	int64
1	year	1922 non-null	int64
2	rank	1922 non-null	int64
3	overall_pick	1922 non-null	int64
4	team	1922 non-null	object
5	player	1922 non-null	object
6	college	1585 non-null	object
7	years_active	1669 non-null	float64
8	games	1669 non-null	float64
9	minutes_played	1669 non-null	float64
10	points	1669 non-null	float64
11	total_rebounds	1669 non-null	float64
12	assists	1669 non-null	float64
13	field_goal_percentage	1665 non-null	float64
14	3_point_percentage	1545 non-null	float64
15	free_throw_percentage	1633 non-null	float64
16	average_minutes_played	1669 non-null	float64
17	points_per_game	1669 non-null	float64
18	average_total_rebounds	1669 non-null	float64
19	average_assists	1669 non-null	float64
20	win_shares	1669 non-null	float64
21	win_shares_per_48_minutes	1668 non-null	float64
22	box_plus_minus	1668 non-null	float64
23	value_over_replacement	1669 non-null	float64

Data Description and Preprocessing

Feature type:

- Numeric (counts, rates)
- Categorical labels
- Advanced metrics

Dropped 246 players with no games played

Final dataset size: 1616 players

Dropped 7 players with no games played

Final dataset size: 53 players

Box Plus/Minus Regression		
Variable	Coefficient	
	Position 1.0 (PG)	Position 5.0 (C)
Pts (adjusted for team context)	0.860	
3PM (3 pointer made)	0.389	
AST	0.580	1.034
TO	-0.964	
ORB	0.613	0.181
DRB	0.116	0.181
STL	1.369	1.008
BLK	1.327	0.703
PF	-0.367	
Variable	Coefficient	
	Offensive Role 1.0 (Creator)	Offensive Role 5.0 (Receiver)
FGA	-0.560	-0.780
FTA (=0.44*FGA)	-0.246	-0.343

So, to calculate **VORP**, the formula is simply: $[BPM - (-2.0)] * (\% \text{ of possessions played}) * (\text{team games}/82)$.

WIN SHARES PER 48 MINUTES

$$WS/48 = 48 \times (\text{Win Shares} / \text{Minutes Played})$$

MARGINAL OFFENSE

$$(\text{points produced}) - 0.92 \times (\text{league points per possession}) \times (\text{offensive possessions})$$

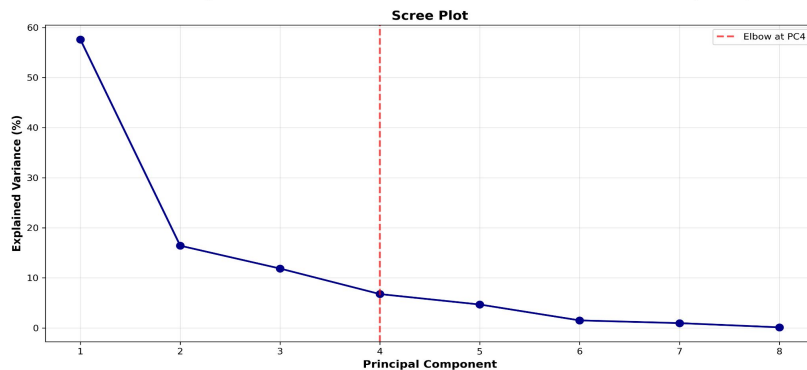
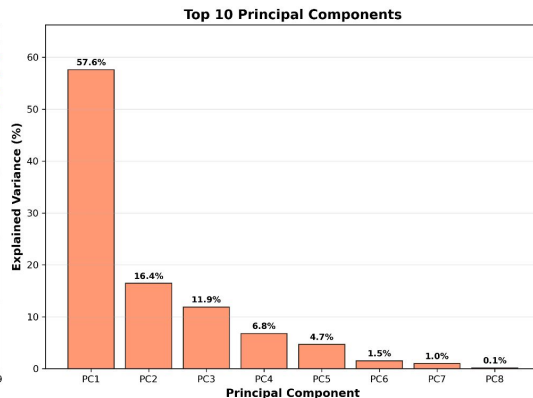
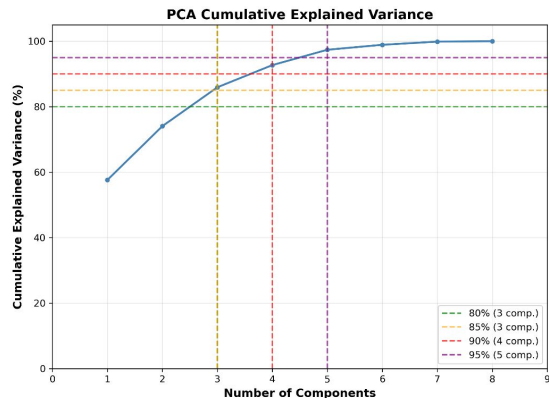
MARGINAL POINTS PER WIN

$$0.32 \times (\text{league points per game}) \times ((\text{team pace}) / (\text{league pace}))$$

OFFENSIVE WIN SHARE

$$(\text{marginal offense}) / (\text{marginal points per win})$$

Exploratory Data Analysis - PCA Tuning



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PC1 (explains 57.62% variance)

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Top 10 features:

1. minutes_played	+ 0.4583
2. points	+ 0.4473
3. games	+ 0.4431
4. total_rebounds	+ 0.4030
5. assists	+ 0.3907
6. free_throw_percentage	+ 0.1806
7. 3_point_percentage	+ 0.1648
8. field_goal_percentage	+ 0.1369

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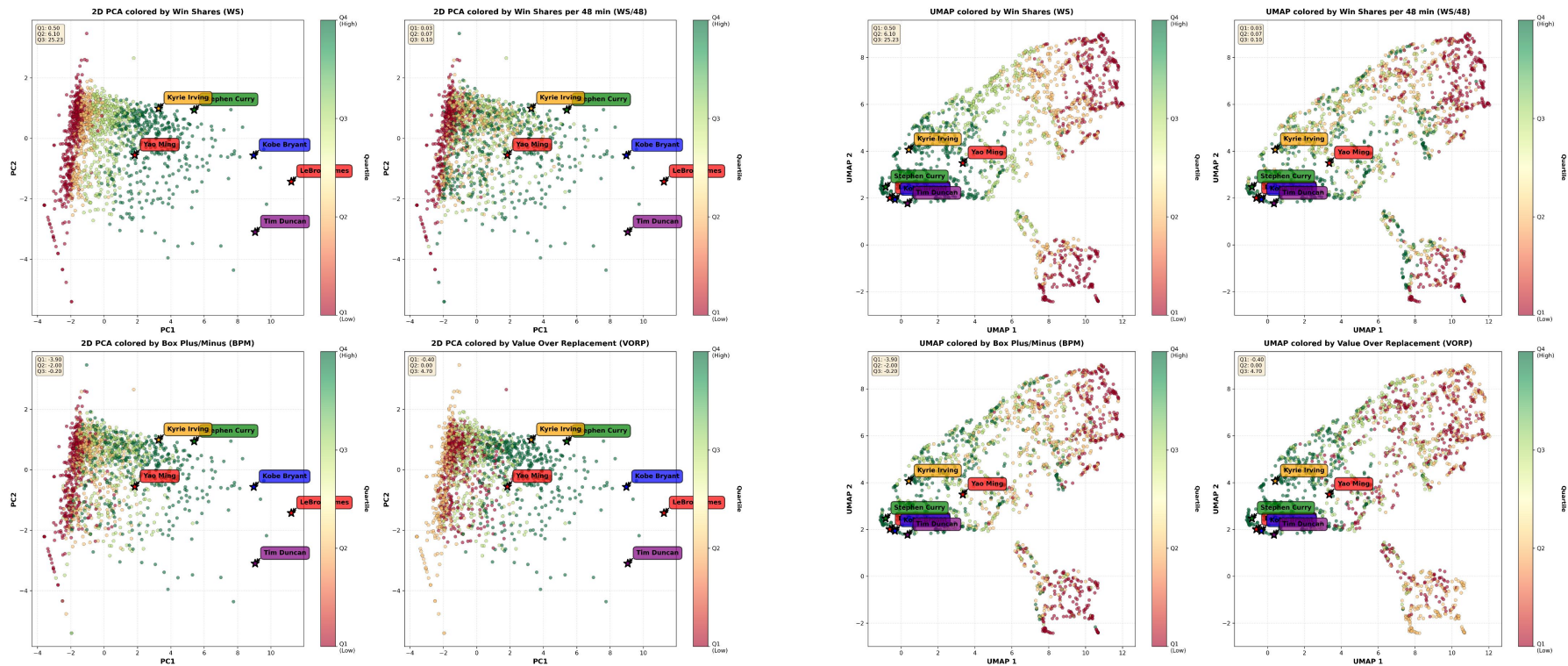
PC2 (explains 16.43% variance)

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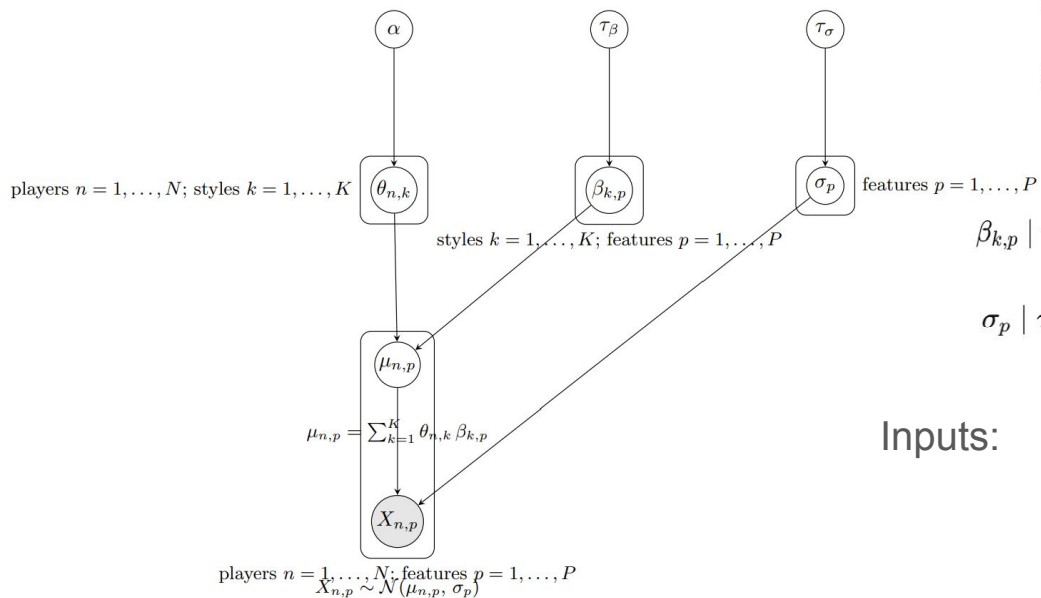
Top 10 features:

1. 3_point_percentage	+ 0.6765
2. free_throw_percentage	+ 0.6155
3. total_rebounds	- 0.2715
4. field_goal_percentage	- 0.2679
5. games	- 0.0799
6. minutes_played	- 0.0775
7. points	- 0.0565
8. assists	+ 0.0503

Exploratory Data Analysis - PCA and UMAP



Model Setup



$\theta_n = (\theta_{n,1}, \dots, \theta_{n,K})$ lives on the K -simplex.

$\theta_n \mid \alpha \sim \text{Dirichlet}(\alpha \mathbf{1}_K), \quad n = 1, \dots, N,$

$\beta_{k,p} \mid \tau_\beta \sim \mathcal{N}(0, \tau_\beta^2), \quad k = 1, \dots, K, p = 1, \dots, P.$

$\sigma_p \mid \tau_\sigma \sim \text{Half-Cauchy}(0, \tau_\sigma), \quad p = 1, \dots, P.$

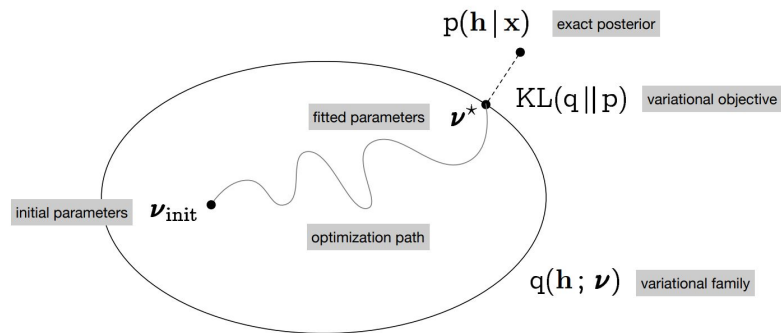
Inputs:

"PTS per 36",
 "REB per 36",
 "AST per 36",
 "Avg minutes",
 "FG% (logit)",
 "3P% (logit)",
 "FT% (logit)",
 "WS/48",
 "BPM",
 "VORP",

Outputs:

$p(\theta, \beta, \sigma, \mu \mid X)$

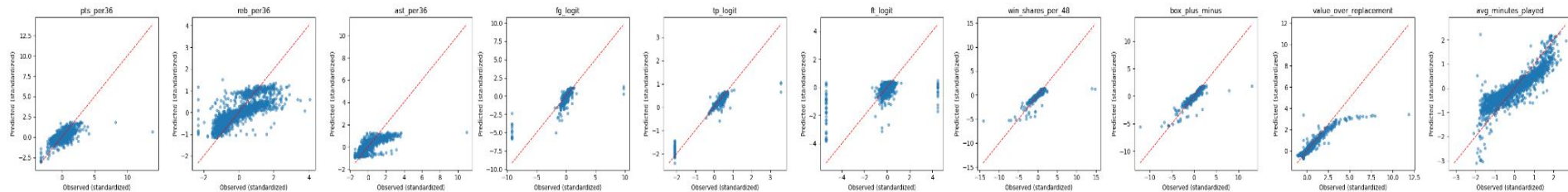
Model Training and Evaluation



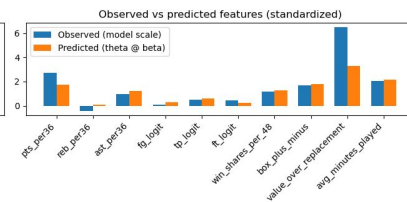
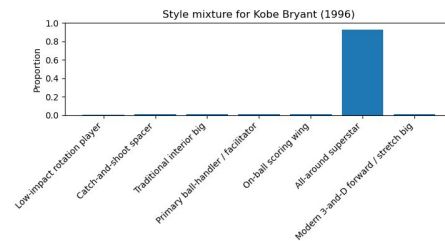
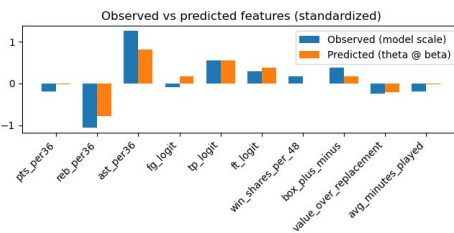
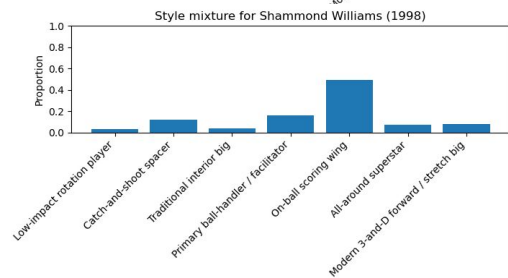
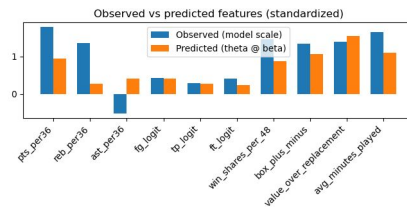
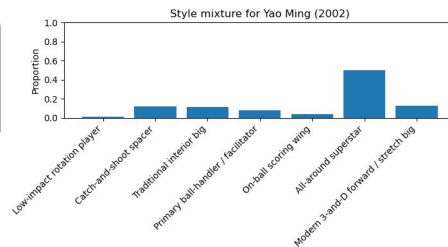
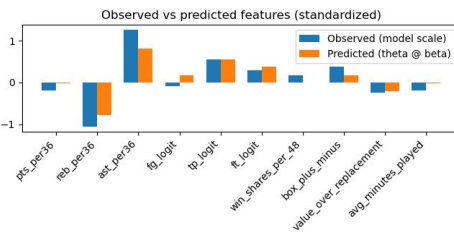
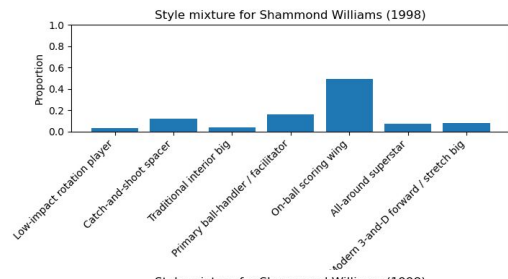
Cross-validation Results for Gaussian Style-Topic Model

K	alpha	Mean MSE	Std MSE	Rank
3	0.5	1.0778	0.0884	2
5	0.5	1.1330	0.0487	3
7	0.5	1.0741	0.0314	1

Reconstruction Loss:

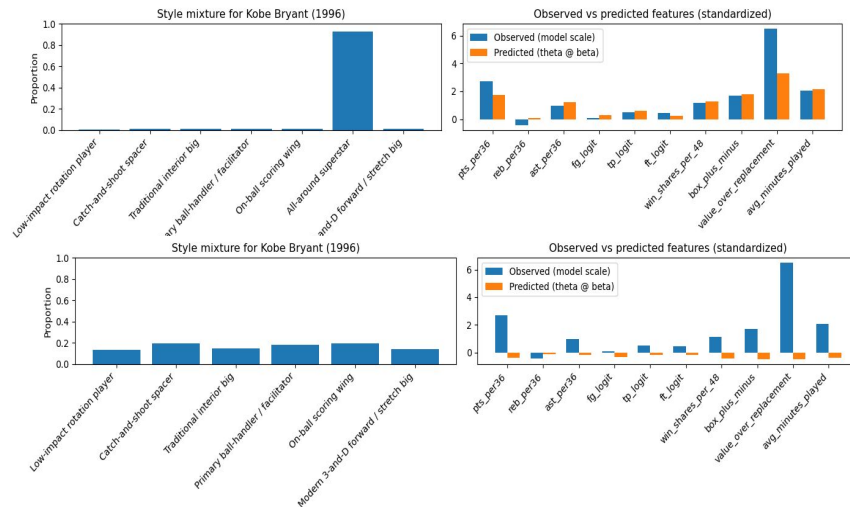


Domain Performance of Selected Model



To be continued: Idea and Challenge

- Degenerate to trivial clustering outcomes, which does not show different patterns between superstars.
- Try to avoid collapse by dropping the most explained principle component. (I'd like to get feedback or suggestion from you.)



For Project details, please refer to our github repository:

<https://github.com/Schuyn/Unsupervised-Machine-Learning-Final-Project>